



Basic Differentiation Rules

D1 $\frac{d}{dx}(c) = 0$ - constant rule

D2 $\frac{d}{dx}(x) = 1$ - identity function rule

D3 $\frac{d}{dx}(cu) = c \frac{du}{dx}$

D4 $\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$ - sum rule

D5 $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$ - product rule

D6 $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$ - quotient rule

D7 $\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$ - power rule

D8 $\frac{d}{dx}(\sqrt{u}) = \frac{\frac{du}{dx}}{2\sqrt{u}}$

D9 $\frac{d}{dx}\left(\frac{1}{u^n}\right) = \frac{-n}{u^{n+1}} \frac{du}{dx}$

Chain Rule

D10 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Inverse Function Rule

D11 $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$

Differentiation of Trigonometric Functions

D12 $\frac{d}{dx}(\sin u) = \cos u \frac{du}{dx}$

D13 $\frac{d}{dx}(\cos u) = -\sin u \frac{du}{dx}$

D14 $\frac{d}{dx}(\tan u) = \sec^2 u \frac{du}{dx}$

D15 $\frac{d}{dx}(\cot u) = -\csc^2 u \frac{du}{dx}$

D16 $\frac{d}{dx}(\sec u) = \sec u \tan u \frac{du}{dx}$

D17 $\frac{d}{dx}(\csc u) = -\csc u \cot u \frac{du}{dx}$

Differentiation of Inverse Trigonometric Functions

D18 $\frac{d}{dx}(\arcsin u) = \frac{1}{\sqrt{1-u^2}} \frac{du}{dx}$

D19 $\frac{d}{dx}(\arccos u) = \frac{-1}{\sqrt{1-u^2}} \frac{du}{dx}$

D20 $\frac{d}{dx}(\arctan u) = \frac{1}{1+u^2} \frac{du}{dx}$

D21 $\frac{d}{dx}(\text{arc cot } u) = \frac{-1}{1+u^2} \frac{du}{dx}$

D22 $\frac{d}{dx}(\text{arc sec } u) = \frac{1}{u\sqrt{u^2-1}} \frac{du}{dx}$

D23 $\frac{d}{dx}(\text{arc csc } u) = \frac{-1}{u\sqrt{u^2-1}} \frac{du}{dx}$



Differentiation of Logarithmic Functions

$$D24 \quad \frac{d}{dx}(\log_b u) = \frac{1}{u} \log_b e \cdot \frac{du}{dx}$$

$$D25 \quad \frac{d}{dx}(\ln u) = \frac{1}{u} \cdot \frac{du}{dx}$$

$$D26 \quad \frac{d}{dx}(u^v) = u^v \cdot \frac{d}{dx}(v \ln u)$$

Differentiation of Exponential Functions

$$D27 \quad \frac{d}{dx}(a^u) = a^u (\ln a) \cdot \frac{du}{dx}$$

$$D28 \quad \frac{d}{dx}(e^u) = e^u \cdot \frac{du}{dx}$$

Differentiation of Hyperbolic Functions

$$D29 \quad \frac{d}{dx}(\sinh u) = \cosh u \frac{du}{dx}$$

$$D30 \quad \frac{d}{dx}(\cosh u) = \sinh u \frac{du}{dx}$$

$$D31 \quad \frac{d}{dx}(\tanh u) = \operatorname{sech}^2 u \frac{du}{dx}$$

$$D32 \quad \frac{d}{dx}(\coth u) = -\operatorname{csch}^2 u \frac{du}{dx}$$

$$D33 \quad \frac{d}{dx}(\operatorname{sech} u) = -\operatorname{sech} u \tanh u \frac{du}{dx}$$

$$D34 \quad \frac{d}{dx}(\operatorname{csch} u) = -\operatorname{csch} u \coth u \frac{du}{dx}$$

Differentiation of Inverse Hyperbolic Functions

$$D35 \quad \frac{d}{dx}(\sinh^{-1} u) = \frac{1}{\sqrt{u^2 + 1}} \frac{du}{dx}$$

$$D36 \quad \frac{d}{dx}(\cosh^{-1} u) = \frac{1}{\sqrt{u^2 - 1}} \frac{du}{dx}$$

$$D37 \quad \frac{d}{dx}(\tanh^{-1} u) = \frac{1}{1 - u^2} \frac{du}{dx}$$

$$D38 \quad \frac{d}{dx}(\coth^{-1} u) = \frac{-1}{1 - u^2} \frac{du}{dx}$$

$$D39 \quad \frac{d}{dx}(\operatorname{sech}^{-1} u) = \frac{-1}{u\sqrt{1 - u^2}} \frac{du}{dx}$$

$$D40 \quad \frac{d}{dx}(\operatorname{csch}^{-1} u) = \frac{-1}{u\sqrt{1 + u^2}} \frac{du}{dx}$$

Hyperbolic Functions are functions which contain a combination of the exponential functions e^x and e^{-x}

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\operatorname{csch} x = \frac{1}{\sinh x}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\operatorname{sech} x = \frac{1}{\cosh x}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\coth x = \frac{\cosh x}{\sinh x}$$



Ratio Identities

$$\tan u = \frac{\sin u}{\cos u} = \frac{\sec u}{\csc u}$$
$$\cot u = \frac{\cos u}{\sin u} = \frac{\csc u}{\sec u}$$

Reciprocal Identities

$$\cot u = \frac{1}{\tan u} \quad \tan u = \frac{1}{\cot u}$$
$$\sec u = \frac{1}{\cos u} \quad \cos u = \frac{1}{\sec u}$$
$$\csc u = \frac{1}{\sin u} \quad \sin u = \frac{1}{\csc u}$$

Pythagorean Identities

$$\sin^2 u + \cos^2 u = 1$$
$$\tan^2 u + 1 = \sec^2 u$$
$$1 + \cot^2 u = \csc^2 u$$

Power Reducing Identities

$$\sin^2 u = \frac{1 - \cos 2u}{2}$$
$$\cos^2 u = \frac{1 + \cos 2u}{2}$$
$$\tan^2 u = \frac{1 - \cos 2u}{1 + \cos 2u}$$

Sum and Difference Identities

$$\sin(u + v) = \sin u \cos v + \cos u \sin v$$
$$\sin(u - v) = \sin u \cos v - \cos u \sin v$$
$$\cos(u + v) = \cos u \cos v - \sin u \sin v$$
$$\cos(u - v) = \cos u \cos v + \sin u \sin v$$
$$\tan(u + v) = \frac{\tan u + \tan v}{1 - \tan u \tan v}$$
$$\tan(u - v) = \frac{\tan u - \tan v}{1 + \tan u \tan v}$$

Double-Angle Identities

$$\sin 2u = 2 \sin u \cos u$$
$$\cos 2u = \cos^2 u - \sin^2 u$$
$$\cos 2u = 2 \cos^2 u - 1$$
$$\cos 2u = 1 - 2 \sin^2 u$$
$$\tan 2u = \frac{2 \tan u}{1 - \tan^2 u}$$

Half-Angle Identities

$$\sin \frac{u}{2} = \pm \sqrt{\frac{1 - \cos u}{2}}$$
$$\cos \frac{u}{2} = \pm \sqrt{\frac{1 + \cos u}{2}}$$
$$\tan \frac{u}{2} = \pm \sqrt{\frac{1 - \cos u}{1 + \cos u}} = \frac{\sin u}{1 + \cos u} = \frac{1 - \cos u}{\sin u}$$



Product to Sum Identities

$$\sin u \cos v = \frac{1}{2} [\sin(u+v) + \sin(u-v)]$$

$$\cos u \sin v = \frac{1}{2} [\sin(u+v) - \sin(u-v)]$$

$$\cos u \cos v = \frac{1}{2} [\cos(u+v) + \cos(u-v)]$$

$$\sin u \sin v = \frac{1}{2} [\cos(u-v) - \cos(u+v)]$$

Sum to Product Identities

$$\sin u + \sin v = 2 \sin\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right)$$

$$\sin u - \sin v = 2 \cos\left(\frac{u+v}{2}\right) \sin\left(\frac{u-v}{2}\right)$$

$$\cos u + \cos v = 2 \cos\left(\frac{u+v}{2}\right) \cos\left(\frac{u-v}{2}\right)$$

$$\cos u - \cos v = -2 \sin\left(\frac{u+v}{2}\right) \sin\left(\frac{u-v}{2}\right)$$

Special Product Formulas

$$(ax + by)(cx + dy) = abx^2 + (ad + bc)xy + bdy^2$$

$$(x + y)^2 = x^2 + 2xy + y^2$$

$$(x - y)^2 = x^2 - 2xy + y^2$$

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x - y)^3 = x^3 - 3x^2y + 3xy^2 - y^3$$

$$(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2xz + 2yz$$

$$(x + y)(x - y) = x^2 - y^2$$

$$(x + y)(x^2 - xy + y^2) = x^3 + y^3$$

$$(x - y)(x^2 + xy + y^2) = x^3 - y^3$$

Laws of Exponents

$$a^m \cdot a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(ab)^n = a^n b^n$$

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

$$(a^m)^n = a^{mn}$$

$$a^{-n} = \frac{1}{a^n}$$



Laws of Logarithmic

$$\log_a a = 1$$

$$\log_a 1 = 0$$

$$\log_b ab = \log_b a + \log_b b$$

$$\log_b \frac{a}{b} = \log_b a - \log_b b$$

$$\log_b a^n = n \log_b a$$

$$\log_b a = \frac{\log a}{\log b}$$

Laws of Radicals

$$(\sqrt[n]{a})^n = a$$

$$\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$\sqrt[m]{a^n} = (\sqrt[m]{a})^n$$

$$\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$